Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	BALANCED
Model	llama3.2

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Editorial Decision

MINOR REVISION

Weighted Score: 3.5 / 5.0
Confidence: 79%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**4.0**	Good	■■■■■
Manuscript Structure	**4.0**	Good	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**3.5**	Good	■■■■■
Results & Evidence	**4.0**	Good	■■■■■
Discussion & Conclusions	**4.0**	Good	■■■■■
Figures & Tables	**0.8**	Very Weak	■■■■■
Scientific Writing	**3.7**	Good	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**3.5**	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **4.0 / 5.0** | Confidence: 90%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Weaknesses:

- [TITLE] could be more concise without losing informativeness.

Major Comments:

- [ABSTRACT] should not contain references as it is supposed to be a standalone summary.

StructureReviewer

Manuscript structure and logical coherence

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a clear research problem and explicitly declares the research question.
- The structure flow is logical, with each section flowing into the next.

Weaknesses:

- Objectives are not clearly stated in the introduction and later addressed in conclusions without explicit connection.
- Section VI is identical to Section V, which may cause confusion.

Recommendations:

- Clearly state objectives in the introduction and explicitly address them in conclusions.
- Consider reorganizing or merging sections V and VI for better coherence.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- clearly defined problem statement
- well-posed theoretical framework
- descriptive experimental setup

Major Comments:

- The logical flow is generally good, but the sequence could be improved by explicitly stating the relationship between the proposed method and the experiments.

Recommendations:

- Consider adding more detail on how the physical admissibility constraints are integrated into the neural mapping.
- Provide more information on the dataset used for validation, including its size and preprocessing steps.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- Bootstrap confidence intervals used with sufficient resamples
- Holm correction applied correctly
- RMSE, MAE, R^2 metrics used and interpreted appropriately
- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Major Comments:

- The use of Kruskal-Wallis test may not be appropriate for the comparison due to non-normal distribution of data.
- Post-hoc corrections (Holm) are applied correctly, but it is unclear if this justifies the choice of test.

Recommendations:

- Consider using non-parametric tests or transformations to address non-normality
- Verify independence assumption for Wilcoxon/rank-sum post-hoc test

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FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **0.8 / 5.0** | Confidence: 90%

Weaknesses:

- Some figure captions could be more descriptive.

Major Comments:

- The figures seem to support the claims made in the text, but it would be beneficial to include more context or explanations for readers who are not familiar with the specific methods or algorithms used.

Recommendations:

- Consider adding unit labels to figure axes where applicable.

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a clear description of the experimental setup and the proposed physics-guided EANN framework.

Weaknesses:

- Some results, such as Pareto analysis and hypervolume/spread indicators, are not clearly explained or justified in the text.

Major Comments:

- The manuscript could benefit from more detailed explanations of the experimental setup and the proposed optimization algorithms.

Recommendations:

- Provide a clear explanation of the Pareto analysis and hypervolume/spread indicators used in the study.
- Include more details about the experimental setup, such as the specific parameters used for data preprocessing and the number of independent runs performed.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.0 / 5.0** | Confidence: 80%

Weaknesses:

- The discussion could benefit from more explicit comparison with prior work in the field of structural parameter identification.

Major Comments:

- While the manuscript presents a well-defined problem and a clear solution, the discussion could delve deeper into the theoretical implications of the proposed EANN framework.
- The experimental validation is convincing, but it would be beneficial to discuss potential limitations and future directions in more detail.

Recommendations:

- Consider adding more explicit comparisons with prior work in the field to strengthen the discussion section.
- Elaborate on potential limitations of the proposed EANN framework and suggest avenues for future research.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.7 / 5.0** | Confidence: 80%

Weaknesses:

- This paper proposes a novel approach to structural parameter identification using physics-guided EANN.
- Paragraphs that are single sentences, e.g., 'The experimental results reported in Section V establish that reliable structural identification under accelerometer-only sensing is achieved.'

Major Comments:

- The manuscript could benefit from more detailed explanations of the physics-guided EANN framework and its application to structural parameter identification.

Recommendations:

- Rephrase sentences to avoid passive voice, e.g., 'The proposed physics-guided EANN framework accurately characterizes the dynamic behavior of the system.'
- Combine single sentences into paragraphs with a clear topic sentence, development, and transition.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 90%

Weaknesses:

- Some references may not directly support the claims made in the manuscript, particularly those that are more recent or less well-known in the field.

Major Comments:

- The references section could benefit from a more balanced mix of seminal works and recent research to strengthen the manuscript's arguments.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Consider adding seminal works from earlier periods, such as the work of L. Mead or R. E. Roberson, to provide a more comprehensive literature review.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] [ABSTRACT] should not contain references as it is supposed to be a standalone summary.

[2] The logical flow is generally good, but the sequence could be improved by explicitly stating the relationship between the proposed method and the experiments.

[3] The use of Kruskal-Wallis test may not be appropriate for the comparison due to non-normal distribution of data.

[4] Post-hoc corrections (Holm) are applied correctly, but it is unclear if this justifies the choice of test.

[5] The figures seem to support the claims made in the text, but it would be beneficial to include more context or explanations for readers who are not familiar with the specific methods or algorithms used.

[6] The manuscript could benefit from more detailed explanations of the experimental setup and the proposed optimization algorithms.

[7] While the manuscript presents a well-defined problem and a clear solution, the discussion could delve deeper into the theoretical implications of the proposed EANN framework.

[8] The experimental validation is convincing, but it would be beneficial to discuss potential limitations and future directions in more detail.

[9] The manuscript could benefit from more detailed explanations of the physics-guided EANN framework and its application to structural parameter identification.

[10] The references section could benefit from a more balanced mix of seminal works and recent research to strengthen the manuscript's arguments.

[11] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Clearly state objectives in the introduction and explicitly address them in conclusions.
2. Consider reorganizing or merging sections V and VI for better coherence.
3. Consider adding more detail on how the physical admissibility constraints are integrated into the neural mapping.
4. Provide more information on the dataset used for validation, including its size and preprocessing steps.
5. Consider using non-parametric tests or transformations to address non-normality
6. Verify independence assumption for Wilcoxon/rank-sum post-hoc test
7. Consider adding unit labels to figure axes where applicable.
8. Provide a clear explanation of the Pareto analysis and hypervolume/spread indicators used in the study.
9. Include more details about the experimental setup, such as the specific parameters used for data preprocessing and the number of independent runs performed.
10. Consider adding more explicit comparisons with prior work in the field to strengthen the discussion section.
11. Elaborate on potential limitations of the proposed EANN framework and suggest avenues for future research.
12. Rephrase sentences to avoid passive voice, e.g., 'The proposed physics-guided EANN framework accurately characterizes the dynamic behavior of the system.'